

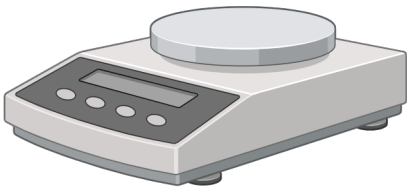
Experimental infections



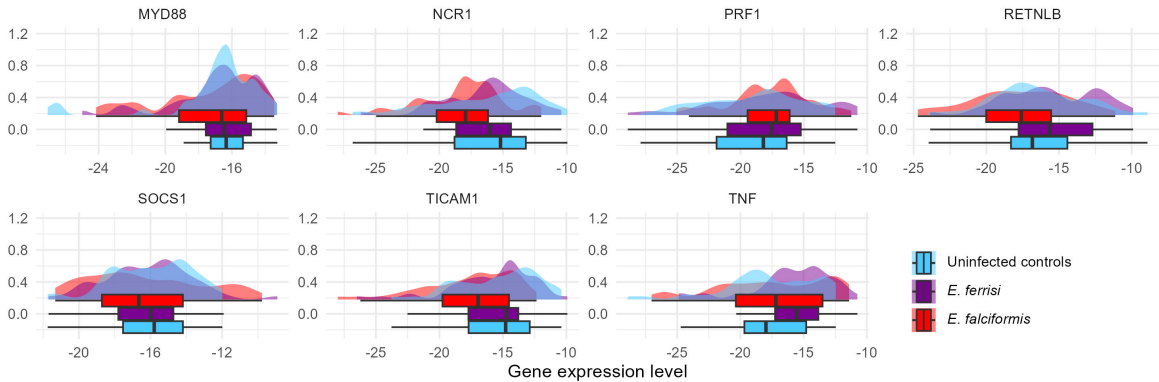
Wild-derived mice strains infected with two strains of *Eimeria* spp.

n = 136

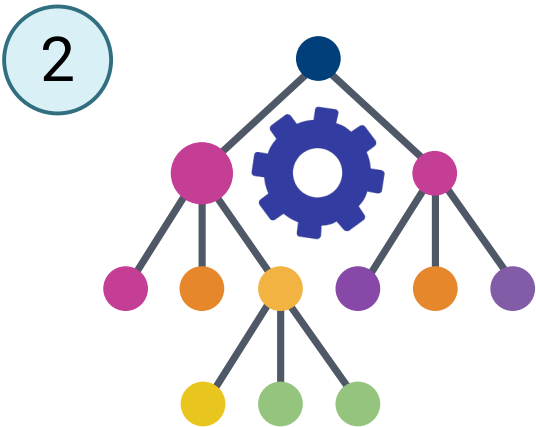
Daily weight measurements:
Proxy for host health



Expression of 19 immune genes
Day 8 post infection



Random forest training

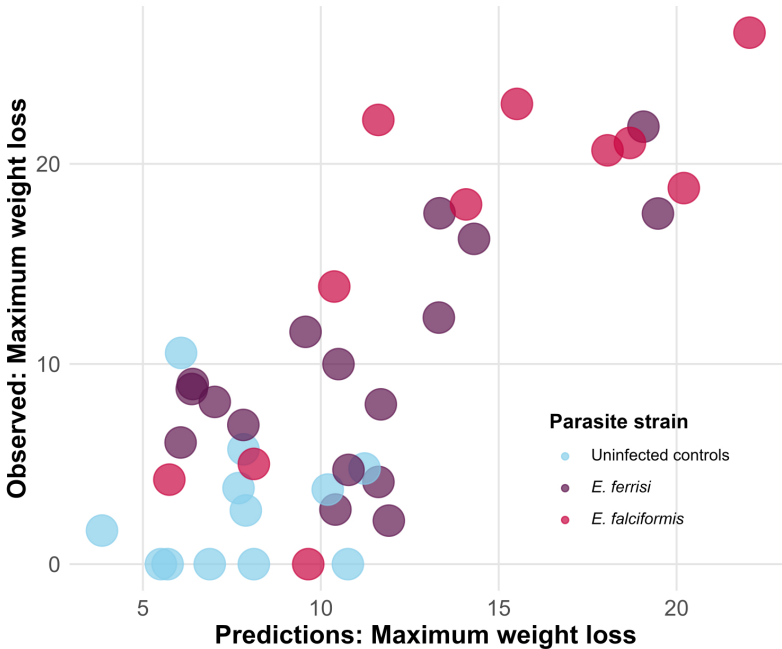


Training data:
Laboratory infection experiments

Independent variables:
19 immune genes

Dependent variable:
Relative weight loss

$r = 0.79, p < 0.0001$
43.6% variance explained

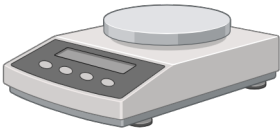


Application:

3 355 wild mice caught in rural Germany



Predicted weight loss for
each individual as a proxy for
host health



Validation of
predictions



Predicted values correlated with
infection intensity ($r = 0.206, p = 0.006$) and *Eimeria* presence in
wild mice